

## Fascial Fitness

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### How to build a youthful, resilient fascial body

The elegant movements of a dancer, the impressive performance of a circus artist, the powerful throw of an Olympic javelin thrower, are not only a matter of muscular strength, good cardiovascular condition, neuromuscular coordination (Jenkins, 2005) and good luck in genetics. According to current findings in the international field of fascia research, the muscular connective tissues, called myofasciae, have more meaning for a body in motion than had been generally considered several decades ago. Recent research findings prove that the body-wide fascial network plays a significant role in force transmission, hydration (fluid dynamic) and proprioception (Chapters 9, 12, 13 and 15).

Specific training, focusing on the question of how to build a strong and flexible fascial body, could be of great value to athletes, dancers, martial arts students and somatic oriented movement advocates. It could also contribute to better prevention of the wide distribution of fascial injuries in athletes (Wilke et al., 2019). The optimal fascial body is both elastic and resilient, and can, therefore, be relied upon to respond effectively to a variety of challenges and circumstances, thereby providing a high degree of injury prevention (Kjaer et al., 2009) (Chapters 4–6).

This chapter mainly focuses on one specific aspect of Fascial Fitness, the elastic storage

capacity of kinetic energy by the collagenous tissues. How to stimulate the fibroblasts to lay down new collagenous fibers in a healthy and youthful network architecture is explored. The physiological and biomechanical foundations that underlie the training principles which follow are described in Chapters 1–6.

The basic principles of a fascia-focused movement training approach, called Fascial Fitness, were first developed by an international group of scientists and clinicians (including Thomas Myers, James Earls, Wilbour Kelsick, Stephen Mutch, Markus Rossmann, Stefan Dennenmoser, Carina Trippelsdorf and Simone Lindner, in addition to the authors of this chapter as well as other contributors). This happened subsequent to the 2009 Fascia Research Congress, at which the surprising changes in length of aponeurotic fasciae versus muscular sarcomeres during hopping movements were impressively demonstrated by Kawakami and Fukunaga (2006). The training principles of Fascial Fitness, as developed by its founders, were first described by Müller and Schleip (2011) and have continued to evolve since then.

### Three modes of fascia-muscle interaction

Previous descriptions of Fascial Fitness, including the corresponding chapter on this topic in the first edition of this book, mainly emphasized the energy conservation mode, in which the

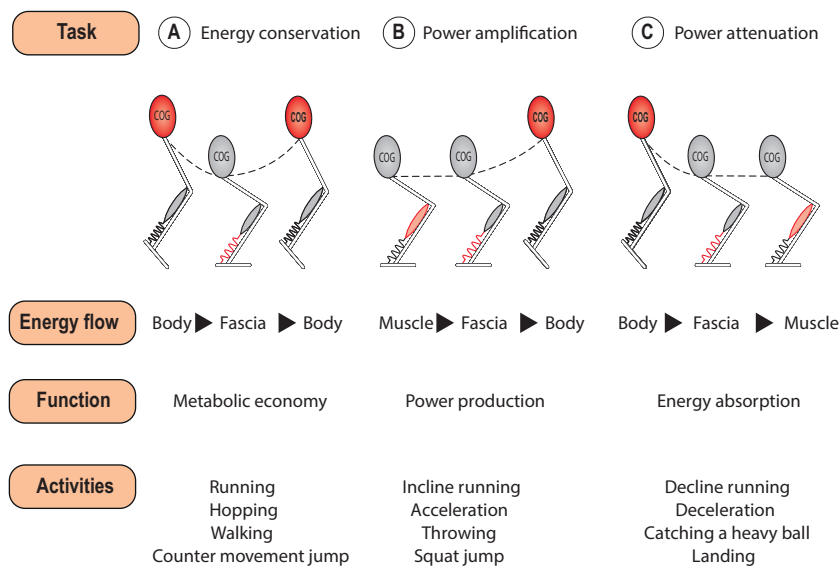
# Chapter 24

strain and subsequent recoil of fascial elements are maximally utilized, as typified by the hopping of a kangaroo (Figure 24.1A). This important mode of fascia-muscle interaction is described in greater detail in Chapter 8.

However, there are two additional modes of functional interaction between fascial elements and muscle fibers that should also be implemented in training routines. Figure 24.1B illustrates the alternative mode of power amplification as used in a single frog jump or in squat jump exercises in plyometrics. Here, a rapid concentric muscular contraction induces a lengthening strain of fascial fibers,

whose subsequent recoil motion then leads to the rapid body dislocation. Regular practice of that modality not only enhances an increase in fascial stiffness, but also in fast-twitch muscle fibers.

Figure 24.1C describes a third mode of fascia-muscle interaction, in which kinetic energy is dampened by lengthening of collagenous fibers as well as by eccentric muscular contraction. Examples of this mode of ‘power attenuation’ include the soft and quiet landing of a gymnast or of a jumping cat on the ground as well as the smooth catching of a flying medicine ball.



**Figure 24.1**

Three modes of functional interaction between parallel fibered fascial tissues (here, mainly tendon and aponeuroses) and muscle. In (A), the energy conservation mode, the center of gravity (COG) shifts down and up, mainly changing the length and tonus of the fascial elements, while the muscle fibers are less strongly involved. In (B), the power amplification mode, a rapid concentric muscular contraction leads the process, which then strains the fascial elements, whose subsequent recoil action then increases the upwards momentum of the COG. In (C), the power attenuation mode, the kinetic energy generated by a drop of COG is absorbed in a dampening fascial elongation, as well as in an orchestrated eccentric muscular contraction. Red color: current storage location of kinetic energy (in terms of an increase in the relative height of COG, or as temporary strain of fascial fibers, or as an active muscular contraction). Illustration modified after Roberts and Azizi (2011) with permission.

The rapid lengthening of both fascial as well as muscular fibers in this application mode tends to go along with more frequent tissue micro-ruptures (or irritations) in both tissues than the other two interaction modes, as seen in a higher incidence of delayed onset muscle soreness (DOMS), which has been shown to include a high sensitization of intrafascial nerve endings (The clinical observation of us authors of the respective fascial length changes during loading suggest that an increase in tendon crimp seems to be particularly expressed in athletes emphasizing mode A, such as runners, whereas an additional increase in epimysial thickness has been observed in athletes emphasizing mode B, such as javelin throwers. In addition, an increase in endomysial thickness has been observed after regular downhill running (mode C) based on yet unpublished work of Prof. Wilhelm Bloch (Univ. of Cologne, Germany). Further research is necessary to clarify the complex architectural tissue adaptations to the three different modes of fascia loading.

## Practical application to enhance power conservation (elastic recoil)

### Preparatory counter-movement

To increase the dynamics of elastic recoil (type A in Figure 24.1), the movement is first initiated with a pre-tensioning in the opposite direction, followed by the actual movement. A suitable metaphor would be an archer who pre-stretches the tendon of their bow in the opposite direction before shooting the arrow in the desired direction. Applying muscular effort in pushing the arrow forward would not be of a comparable effectiveness (Chapter 8).

### Frontal leg

Standing with feet hip-width apart, shift the weight onto one leg (Figure 24.2). At the beginning, to aid balance, hold on to the back of a chair. To progress, you may later remove the balancing aid, once the movement is familiar and fluent.



**Figure 24.2 Leg swing**

(A) The leg is first extended backwards in such a way that a pre-stretch is created in the front.

(B) The stored tension is then suddenly released and the leg is accelerated forward like a swinging pendulum. Alternating between a dorsiflexed foot position (as shown here) and a plantarflexed position will add variation of the loading focus to different portions of the swinging leg.

## Chapter 24

- Start with easy swings of the free leg, swinging backwards and forwards like a pendulum. In such a swinging motion, kinetic energy is rhythmically stored and released; however, the storage of kinetic energy occurs in the spatial relationship of the swinging weight towards gravity, whereas the fascial tissues are not—or not yet—stretch-loaded.
- Increase the loading by deliberately pre-stretching in the opposite direction (backwards), followed by releasing the stored energy through the frontal swing. Here, the kinetic energy is also stored and released, although this time the storage involves an elastic elongation (stretch) of collagenous tissues within the body.
- To further enhance the elastic recoil effect, initiate the frontal leg swing proximally, from your pubic bone or sternum, immediately followed by the distal end, via the pre-stretched leg and foot.
- To load the tissues even more effectively and enhance proprioceptive refinement, use ankle weights.

### Fascial effects

The frontal leg swings are optimal to increase elasticity in short hip flexors and to lengthen hamstrings.

### Fascial Fitness: Basic elastic recoil exercise – Flying Sword

The Fascial Fitness exercise called the Flying Sword is one of the core exercises to train elastic recoil, especially in the lumbodorsal fascia. Beginners start with steps 1 to 3, subsequently adopting the more refined aspects presented in steps 4 and 5 once the earlier steps have been mastered. Most important is the orchestration

of the movement, without muscular effort or strain, and optimally performed as fast, fluid and powerful.

### Three basic steps

1. Preparatory counter-movement
2. Proximal initiation of the power motion
3. Sequential delay of more distal body parts in following this movement.

### Equipment

Weight, dumbbell, kettle bell, swing dumbbell.

Stand with feet a little wider than hip-width apart so that the weight can easily be moved between your knees.

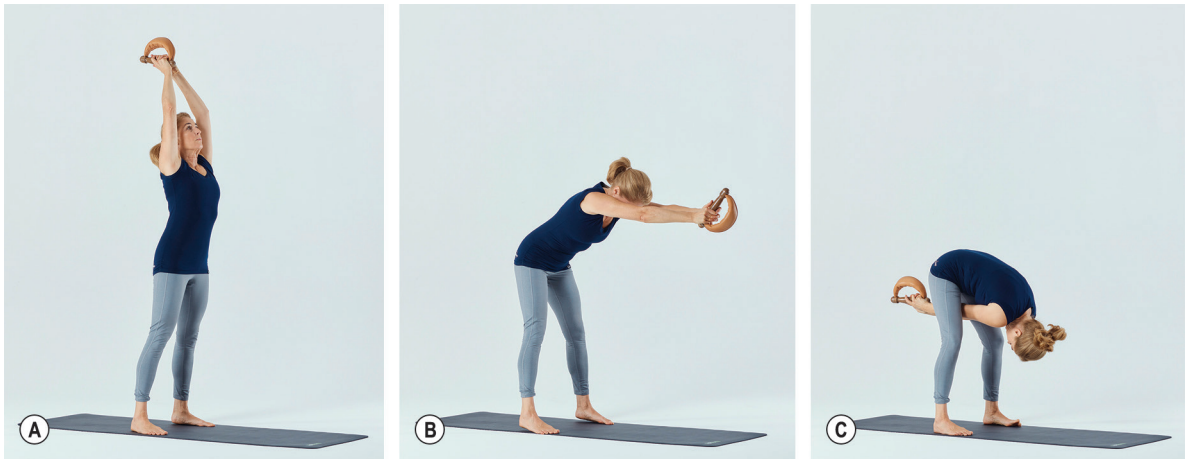
### Step 1: Preparatory counter-movement

Hold the weight with both hands and lift your arms up above your head. Pre-tensioning is achieved, while bending the body's axis slightly backward and extending it in an upward direction, lengthening at the same time (Figure 24.3A). This quick backward bounce initiates the movement and increases the elastic tension in the front fascial “body suit”.

Bring the weight down, by releasing the pre-tension through the upper body and arms. This allows them to spring forward and down like a dynamic catapult, allowing the weight to “fly” like a sword between your knees (Figures 24.3B and 24.3C).

Next, reverse this process. Here, the recoil capacity of the fascia is activated by an active pre-tensioning of the posterior fascia by directing the weight further backwards. Before moving from the forward bending position, the flexor muscles on the front of the body are first, briefly, activated. This momentarily pulls the body even further back and down and at the same time the fascia





**Figure 24.3 Flying Sword**

(A) The principle of preparatory counter-movement is an effective way to load the fascial tissues, before performing the actual movement in the desired direction. For example, in this exercise we start the movement with a pre-stretch into the opposite direction, slightly bending backwards.

(B) Releasing down: the stored energy in the connective tissues allow a dynamic end efficient movement performance. In this phase we “fly” down, with hardly any muscular effort, but relying instead on the capacity of fascia to store kinetic energy and release it.

(C) At the turning point: pull the weight and the upper body slightly more backwards into the preparatory counter-movement. To increase the recoil effect and loading of the fascia of the back, briefly activate the flexor muscles on the front of the body first, before releasing the stored energy in flying up into the starting position.

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on the back of the body is loaded with greater tension. The kinetic energy, which is stored on the posterior side of the fascial net, is dynamically released via a passive recoil effect as the upper body flies back to the original upright position.

### Rhythm is it

A feeling of rhythm is required to ensure that the individual is not relying on the muscle work of their back muscles, but rather on the dynamic recoil activity of the fascia, in addition to the preparatory counter-movement. This is similar to the timing necessary when playing with a yo-yo. If the inherent rhythm is met, it swings with almost effortless ease and flow (Chapter 8).

### Step 2: Proximal initiation to perform the actual motion

From the pre-tensed backward bending position, the forward movement is initiated by a proximal pull of the sternum followed by the distal parts of the body (Figure 24.4). In this exercise, the sternum or pubic bone initiates the release forwards and downwards.

### Step 3: Sequential delay of more distal body parts following this movement

The proximal initiation of Step 2 is immediately followed by a sequential delay of the more distal body parts. In this exercise, the arms and

## Chapter 24



**Figure 24.4 Proximal initiation and distal delay**

To increase the elastic recoil effect and its power in action, we initiate the movement by the proximal pull of the sternum, followed by a distal delay of the upper body, in this example by the arms and the hands, creating a flowing whip-like movement and thereby increasing the energy storage and release.

hands holding the weight follow behind, creating a wave-like movement. By using the proximal initiation and sequential delay of the distal parts, the pre-tension of the body is further increased, and dynamic power and acceleration are enhanced.

### Two advanced steps to enhance elastic recoil

Having mastered these basic steps in a recoil movement exercise, add these two more advanced steps to refine the orchestration of the movement.

The following two steps are practiced immediately prior to the three basic steps described above.

#### Advanced step 1: Increase of sensory awareness – proprioceptive refinement

One of the big surprises resulting from fascial research was the discovery that fascia contains a rich supply of sensory nerves, including proprioceptive receptors, multimodal receptors and nociceptive nerve endings. This means that fascia is alive (Chapter 15)!

Histological findings indicate that the superficial fascial layers of the body are, in fact, much more densely populated with sensory nerve endings than the connective tissues situated internally (Benetazzo et al., 2011; Tesarz et al., 2011). In particular, the transition zone between the fascia profunda and the subdermal loose connective tissue seems to have the highest sensorial innervation (Tesarz et al., 2011). The body-wide connective tissue network is certainly our most important organ for proprioception (Schleip, 2003).

For a long time, proprioception shared a similar history to fascia, and, at the start of the 21st century, was rediscovered as our main movement sense. In a movement training routine, there are a variety of ways to train and stimulate sensory or proprioceptive refinement. Common

and effective ways before the actual movement include rubbing, tapping, rolling, brushing or kinesiotaping specific bodyparts to stimulate the receptors of the superficial fascial layers.

In a more refined preparation, mindfulness and the benefits of conscious attention focus may be used (Moseley et al., 2008). For example, by focusing on the natural expansive wave of inhalation to expand from within into the surrounding space or by sensing the slightest touch of the air on the skin (Chapter 47).

### Advanced step 2: Tensegral expansion

Tensegrity (Chapter 11), and the essential role that fascial membranes play in the structural well-being of the body, are utilized in the principle of tensegral expansion. However, before we move into action, we must first engage a 360-degree body-wide spacious expansion. This can be achieved by an all-over pre-tensioning of the superficial fascia, which envelopes the body as a whole. In Fascial Fitness training, we describe this as “tensing the tiger body suit”. Research findings show that this kind of tensegral pre-tension is a substitute for a “power pose” that is known from the animal kingdom. This has instant positive effects by changing our psychological condition (Carney et al., 2010).

In the Flying Sword exercise, this would be created by bringing attention to the body’s “poles”, engaging a slight bi-directional movement apart, from top to toe; or in fascial terms, from the plantar fascia to the galea aponeurotica, as well as a widening from front to back. The same kind of attention is added while bending backwards, to load the fascial chain of the front. Here, the tensegral expansion between the vertebrae is enhanced, to avoid any buckling and, therefore, straining in the cervicals or the lumbar.

### Practical application to enhance power amplification

In the principle of power amplification (Figure 24.1B), the motion is started with a rapid concentric muscular contraction. This is partially transmitted into a lengthening strain of tendons, aponeuroses and other fascial tissues, whose subsequent elastic recoil action then adds additional momentum to the overall joint motion. It is ideal if the tonicity (pre-stress) of the fascial tissues fits exactly to the expected magnitude of the muscular contraction, that is, for a moderate muscular contraction, a moderate fascial pre-stress may be best suited, while a very powerful overall motion may require a much stronger fascial pre-stress.

Contrary to the previously described elastic recoil motions, such power amplification motions are not initiated with a quick preparatory counter-movement. In the athletic field of plyometrics, many forms of “squat jump” training variations exist, which are well suited to this method of muscle-fascia interaction. Similarly, many athletic throwing techniques, such as javelin throwing, fit well into this basic principle.

### Frog jump

Start in a squatted position, with your knees directly over your front feet. Keep your arms extended backwards and raise your chest slightly in an upward/forward direction. Experiment with finding the ideal bending position for your ankles, knees and hip joints, as well as the best position for your arms, to give you maximum momentum. Do not initiate the jump with a preparatory downwards bounce, but rather start the jumping with an explosive upwards motion.

## Chapter 24

Figure 24.5 shows the more joyous motion variation, in which the arms are allowed to contribute additional momentum. An alternative version can also be conducted without the upwards arm swing. In this version, the power amplification of the legs is more strongly emphasized.

### Practical applications to enhance power attenuation

#### Daily fascial training: Stair dancing

Walking up and down stairs can become an instant fascial recoil training when the Ninja principle (see below) is applied. A variation of easy bounces in stepping is suggested with the focus of making as little noise as possible. The “no sound parameter” provides useful feedback to engage the fascial springiness: the quieter and gentler the better. Dancing fascially up and down the stairs would be of additional benefit using a barefoot-like plantar-foot contact with the ground (Chapter 32).

#### The Ninja principle

The Ninja principle uses the metaphor of the legendary Japanese warriors, who reputedly moved as silently as cats leaving no trace. When engaging fascial elements in bouncy movements, such as jumping, running or dancing, the quality of motion should be as smooth and quiet as possible. A change in direction is preceded by a gradual deceleration of the movement before the turning point and a gradual acceleration afterwards. Any harsh, jerky or noisy movement would be counter-productive. The sensed benefits could be a perception of fluid, elegant and effective movement, like a cat in a dynamic leap or stalking while hunting (Figure 24.6).

#### Fascial stretching

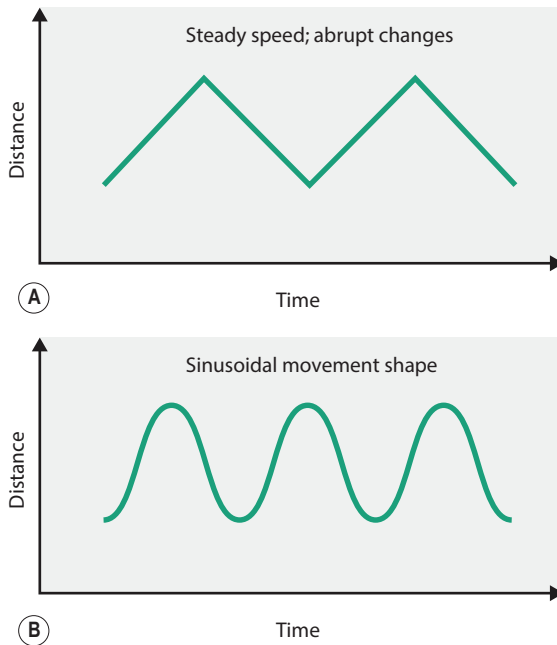
In Fascial Fitness, both dynamic and slow stretching applications are used. Instead of stretching isolated muscle groups, the aim is finding body movements that engage the longest possible myofascial chains (Chapter 13). In the slower



**Figure 24.5** The exercise frog jump demonstrates one of several options for the principle of power amplification

(A) In a squatted position, proper pre-tension is tensegrally distributed throughout the whole body, including the arms.

(B) The release of this pre-tension is then immediately followed by an explosive upwards jumping motion.



**Figure 24.6 Movement quality in change of direction: jerky versus elegant turns**

When a dynamic arm movement (e.g. forward and back), is performed with lack of proprioceptive refinement, the tendency is to use sudden turns and provoke **(A)** abrupt loading patterns. By contrast, when the same movements are conducted with an internal search for elegance, then **(B)** a more sinusoidal movement directional change can be observed. This is characterized by a gradual deceleration before the turning point, which is followed by a subsequent gradual acceleration. In this pattern, the loaded tissues are less prone to injuries, and the movements appear fluid and graceful.

stretches, this is not done by passively waiting, as in a lengthening classical Hatha yoga pose, or in a conventional isolated muscle stretch. Because the majority of the human fascial net is composed of membranous sheets, rather than long narrow stripes, multidirectional angular variations are frequently explored during stretching. This might include sideways or diagonal movement variations as well as spiraling rotations. With this method, large areas of the fascial network are simultaneously involved.

The dynamic stretching variation may be familiar to many readers, as it was part of the physical education in the first half of the 20th century. In recent decades, such bouncing stretches were considered to be generally harmful to the tissue, but contemporary research has confirmed the method's merits. While stretching immediately before athletic performance (e.g. 30 s before a

competitive run) may induce minor attenuations in the performance of world-class athletes, long-term and regular use of dynamic stretching, when correctly performed, can positively influence the architecture of the connective tissue, in that it becomes more elastic and injury resistant (Decoster et al., 2005; Chaabene et al., 2019). For dynamic stretching, the muscles and tissue should be warmed up first, and any jerking or abrupt movements should be avoided. The aforementioned principles of elastic recoil can often be applied, including the proprioceptive refinement, using a sinusoidal deceleration and the preparatory counter-movement.

When the elongated myofibers are in a relaxed condition then a slow stretching approach is sufficient to reach many intramuscular fascial tissues. However, this approach does not reach the tendinous tissues as they are arranged in



## Chapter 24

series with the relaxed and soft myofibers (Figure 24.7C). In order to stimulate these tendinous and aponeurotic tissues, more dynamically swinging stretch movements are recommended, similar to the elegant and fluid extensional movements of rhythmic gymnasts. The same tissues can also be targeted by muscular activation (e.g. against resistance) in a lengthened position, similar to how a cat sometimes enjoys pulling their front claws towards the trunk when stretching (Figure 24.7D). Finally, so-called “mini-bounces” can also be employed as soft and playful explorations in the lengthened stretch position.

### Fascial training guidelines

One main intention of a connective tissue-oriented training routine is to influence the matrix remodeling via specific training activities, which may, after 3–12 months, result in a more resilient “silk-like body suit”. This creates a strong and elastic fibrous network that is at the same time flexible, allowing smooth gliding joint mobility over wide angular ranges. A few Fascial Fitness training guidelines to support optimal results are listed below.

#### Low loading

Inclusion of fascial rebound and long myofascial chains often triggers an exhilarating sense of playfulness, fun and adventure orientation. However, if untamed, this can also lead to more frequent injuries than standard muscle training, which features monotonous repetitions. Start with much lower loads and repetitions than usual. Increase load only if a sense of elegance can be maintained, particularly during the elastic rebound phase.

#### Low frequency

The examination of collagen turnover in a tendon after exercise has shown that collagen synthesis is, indeed, increased after exercise.

Yet the fibroblasts also break collagen down at the same time. Furthermore, 24–48 h after exercise, collagen degradation is greater than collagen synthesis. After 48 h and thereafter this situation is reversed. Therefore, it is suggested that this appropriate tissue stimulation only occurs 1–2 times per week (Magnusson et al., 2010) (Chapter 1).

### Long-lasting effect

In contrast to muscular strength training, in which big gains occur rapidly in only a few weeks, fascia renewal is much slower (Chapter 1). Therefore, the improvements during the first few weeks may be small and less obvious on the outside. It usually takes 3–9 months to see tissue remodeling effects from the outside as well as “to feel” them in palpation. In muscular training, a plateau is reached relatively quickly, with further increases difficult to achieve. However, fascial improvements have a cumulative effect and will not be lost as quickly (e.g. when training is stopped because of health- or work-related reasons), and are, therefore, of a more sustainable quality (Kjaer et al., 2009) (Chapter 5). Regular application over 2–3 years will definitely be expected to yield long-lasting tissue improvements in the form of improved strength and elasticity of the global fascial network.

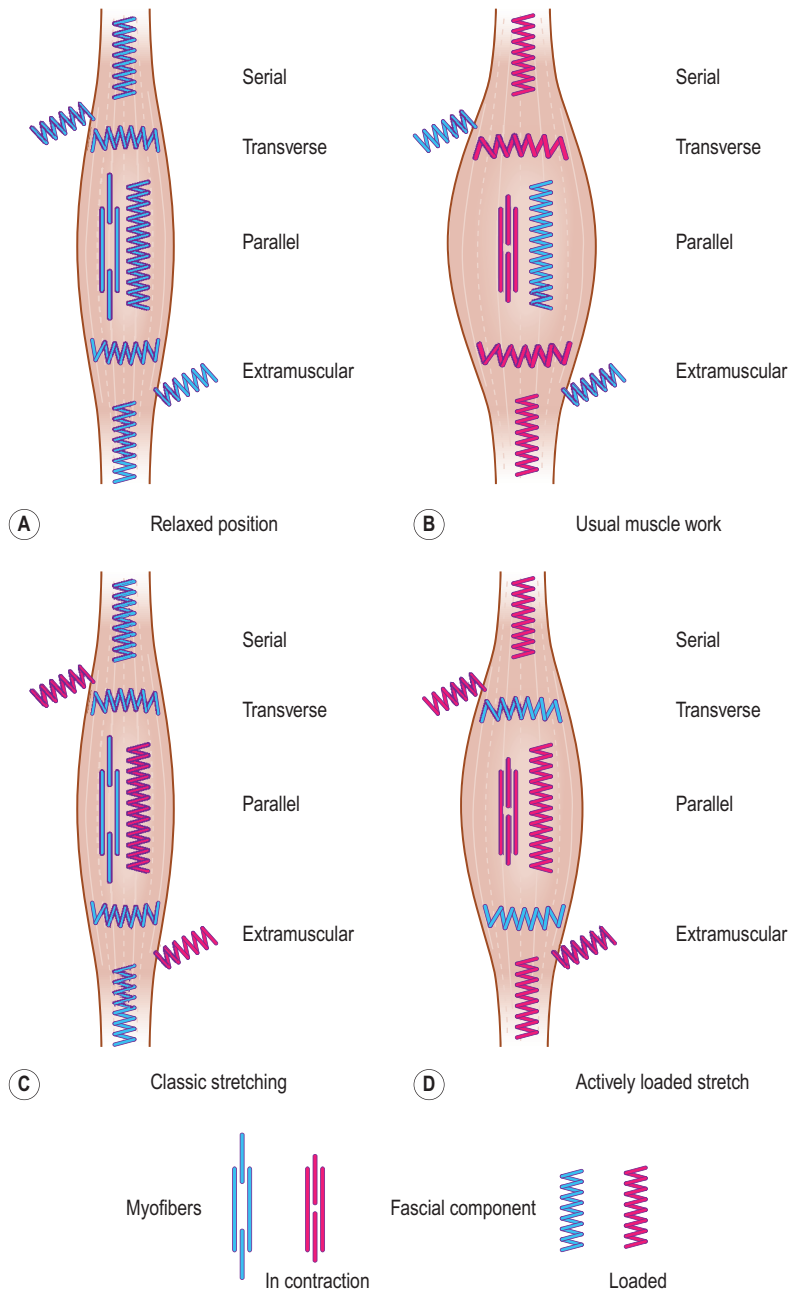
### Clinical summary

Fascial Fitness training does not seek to compete with neuromuscular or cardiovascular training, both of which can have very important health effects that are not possible with fascial training on its own. On the contrary, fascial training is suggested as either a sporadic or more regular addition to comprehensive movement training. It promises to lead towards remodeling of the body-wide fascial network in such a way that it works with increased effectiveness and refinement

# Fascial Fitness

in terms of its kinetic storage capacity, as well as a sensory organ for proprioception. Further research is required to validate whether it does,

indeed, fulfill its basic promise of increased protection against repetitive strain injuries in sports medicine.



**Figure 24.7 Loading of different fascial components**

**(A)** Muscle in relaxed position: the tissue muscle is at normal length and the myofibers are relaxed. Here, none of the fascial elements are being stretched.

**(B)** Usual muscle work: the myofibers are actively contracted while the muscle is at normal length range. Those fascial tissues that are loaded are either arranged in series with the myofibers or run transverse to them.

**(C)** Classic stretching: the muscle is elongated and its contractile myofibers are relaxed. Here, fascial tissues that are being stretched are oriented parallel to the myofibers, as well as extra-muscular connections. However, fascial tissues oriented in series with the myofibers are not sufficiently loaded, since most of the elongation in that serially arranged force chain is taken up by the relaxed myofibers.

**(D)** Actively loaded stretch: the muscle is activated in long end range positions. Most of the fascial components are being stretched and stimulated in that loading pattern.

Note that various mixtures and combinations between the four different fascial components exist. This simplified abstraction therefore serves as a basic orientation only.

## Chapter 24

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